

MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.1 Worksheet

1. Consider the points $A(-4, 0, -1)$, $B(3, 1, -5)$, & $C(2, 4, 6)$.

a) Determine the distance between points A and B.

$$d = \sqrt{(-4-3)^2 + (0-1)^2 + (-1-(-5))^2} = \sqrt{49+1+16} = \sqrt{66}$$

b) Which of these points is closest to the yz -plane?

We can observe the x -value of each point to see how far away it is from being 0 (i.e., being at the yz -plane). After doing this we can see that point C is the closest to the yz -plane.

c) Which point lies in the xz -plane?

We can observe the y -value of each point to see how far away it is from being 0 (i.e., being at the xz -plane). After doing this we can see that point A is the closest to the xz -plane.

d) Which point is closest to the origin?

We will calculate the distance from each point to the origin.

$$d_A = \sqrt{(-4-0)^2 + (0-0)^2 + (-1-0)^2} = \sqrt{16+0+1} = \sqrt{17}$$

$$d_B = \sqrt{(3-0)^2 + (1-0)^2 + (-5-0)^2} = \sqrt{9+1+25} = \sqrt{35}$$

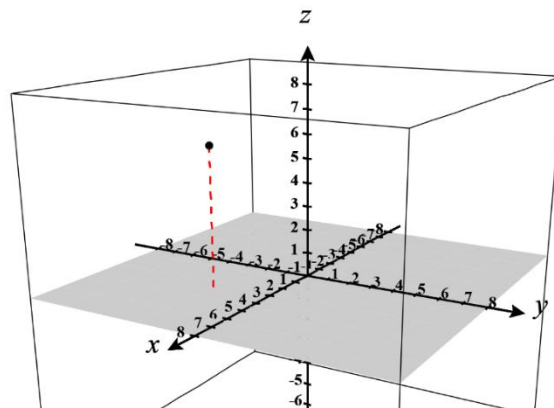
$$d_C = \sqrt{(2-0)^2 + (4-0)^2 + (6-0)^2} = \sqrt{4+16+36} = \sqrt{56}$$

We can see that the point A is closest to the origin.

2. Find the distance from $(4, -2, 6)$ to each of the following.

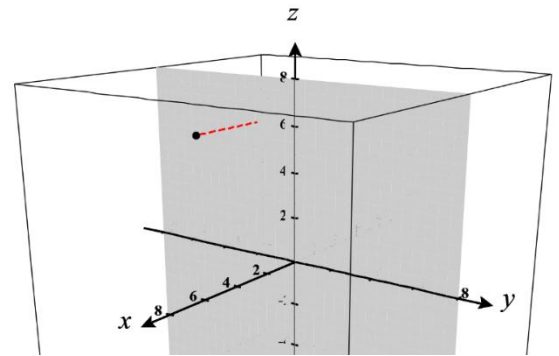
a) The xy -plane

We can observe the z -value of the point to see how far away it is from being 0 (i.e., being at the xy -plane). So this point has a distance of 6 from the xy -plane.



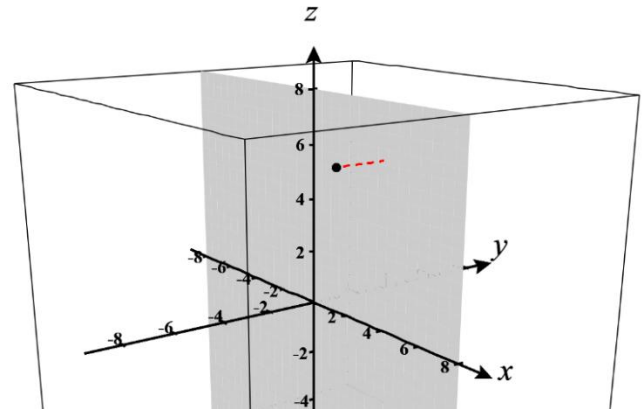
b) The yz -plane

We can observe the x -value of the point to see how far away it is from being 0 (i.e., being at the yz -plane). So this point has a distance of 4 from the yz -plane.



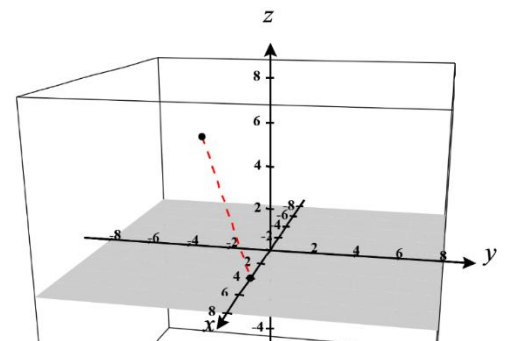
c) The xz -plane

We can observe the y -value of the point to see how far away it is from being 0 (i.e., being at the xz -plane). So this point has a distance of 2 from the xz -plane.



d) The x -axis

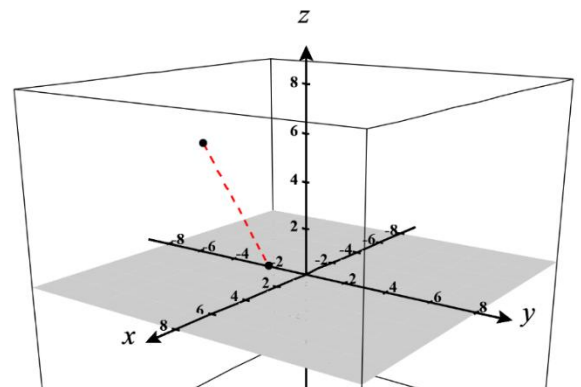
The point closest to $(4, -2, 6)$ on the x -axis would have coordinates $(x, 0, 0)$. Therefore the distance between these points would be $d = \sqrt{(4-x)^2 + (-2-0)^2 + (6-0)^2}$. In order to minimize this distance we would need to set $x = 4$. Therefore, the point on the x -axis that is closest to $(4, -2, 6)$ is $(4, 0, 0)$.



e) The y -axis

The point closest to $(4, -2, 6)$ on the y -axis would have coordinates $(0, y, 0)$. Therefore the distance between these points would be

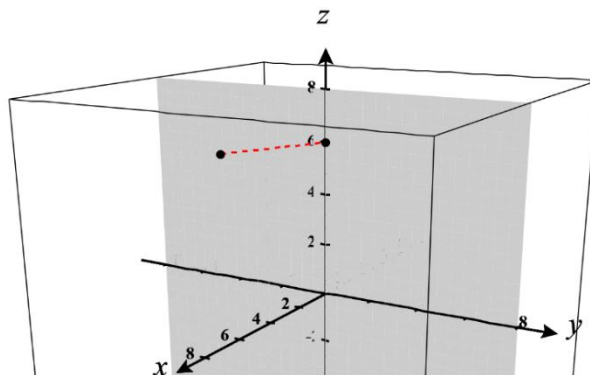
$d = \sqrt{(4-0)^2 + (-2-y)^2 + (6-0)^2}$. In order to minimize this distance we would need to set $y = -2$. Therefore, the point on the y -axis that is closest to $(4, -2, 6)$ is $(0, -2, 0)$.



f) The z -axis

The point closest to $(4, -2, 6)$ on the z -axis would have coordinates $(0, 0, z)$. Therefore the distance between these points would be

$d = \sqrt{(4-0)^2 + (-2-0)^2 + (6-z)^2}$. In order to minimize this distance we would need to set $z = 6$. Therefore, the point on the z -axis that is closest to $(4, -2, 6)$ is $(0, 0, 6)$.



3. Find an equation of a sphere with center $(-3, 2, 5)$ and radius 4. What is the intersection of this sphere with the yz -plane?

The equation of the sphere is

$$(x - (-3))^2 + (y - 2)^2 + (z - 5)^2 = 4^2$$

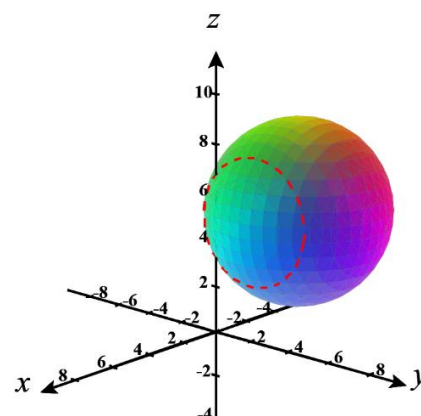
$$\rightarrow (x + 3)^2 + (y - 2)^2 + (z - 5)^2 = 16$$

When this sphere intersects the yz -plane at points with x -coordinates of 0.

To find this intersection we set $x = 0$ in the equation of the sphere. We see that

$$(0 + 3)^2 + (y - 2)^2 + (z - 5)^2 = 16 \rightarrow (y - 2)^2 + (z - 5)^2 = 7$$

This means that the sphere intersects the yz -plane with a circle of points (centered at $(2, 5)$ and with radius $\sqrt{7}$).



4. Find an equation of the sphere that passes through the point $(4, 3, -1)$ and has center $(3, 8, 1)$

We first determine the radius of the sphere by checking the distance between these points.

$$r = \sqrt{(4-3)^2 + (3-8)^2 + (-1-1)^2} = \sqrt{1+25+4} = \sqrt{30}$$

Therefore, the equation of the sphere is: $(x-3)^2 + (y-8)^2 + (z-1)^2 = 30$.

5. Find an equation of the sphere that passes through the origin and whose center is $(1, 2, 3)$

We first determine the radius of the sphere by checking the distance between these points.

$$r = \sqrt{(1-0)^2 + (2-0)^2 + (3-0)^2} = \sqrt{1+4+9} = \sqrt{14}$$

Therefore, the equation of the sphere is: $(x-1)^2 + (y-2)^2 + (z-3)^2 = 14$.

6. Show that the equation represents a sphere, and find its center and radius.

a) $x^2 + y^2 + z^2 - 2x - 4y + 8z = 15$

$$\begin{aligned}(x^2 - 2x + \underline{\quad}) + (y^2 - 4y + \underline{\quad}) + (z^2 + 8z + \underline{\quad}) &= 15 \\(x - 2x + 1) + (y^2 - 4y + 4) + (z^2 + 8z + 16) &= 15 + 1 + 4 + 16 \\(x - 1)^2 + (y - 2)^2 + (z + 4)^2 &= 36\end{aligned}$$

This is a sphere with center $(1, 2, -4)$ and radius 6 .

b) $2x^2 + 2y^2 + 2z^2 = 8x - 24z + 1$

$$\begin{aligned}(2x^2 - 8x + \underline{\quad}) + (2y^2) + (2z^2 + 24z + \underline{\quad}) &= 1 \\2(x^2 - 4x + \underline{\quad}) + 2(y - 0)^2 + 2(z^2 + 12z + \underline{\quad}) &= 1 \\2(x - 4x + 4) + 2(y - 0)^2 + 2(z^2 + 12z + 36) &= 1 + 8 + 72 \\2(x - 2)^2 + 2(y - 0)^2 + 2(z + 6)^2 &= 81 \\(x - 2)^2 + (y - 0)^2 + (z + 6)^2 &= \frac{81}{2}\end{aligned}$$

This is a sphere with center $(2, 0, -6)$ and radius $\frac{9}{\sqrt{2}}$.

7. Prove that midpoint of the line segment from $P_1(x_1, y_1, z_1)$ to $P_2(x_2, y_2, z_2)$ is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$

We know that the distance between P_1 and P_2 is given by $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$. If the point $P_M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$ is the midpoint between P_1 and P_2 , then it should be the case that the distance between P_1 & P_M and P_2 & P_M should both be equal to $\frac{1}{2}d$.

We see that the distance between P_1 & P_M is

$$\begin{aligned}d_M &= \sqrt{\left(x_1 - \frac{x_1 + x_2}{2}\right)^2 + \left(y_1 - \frac{y_1 + y_2}{2}\right)^2 + \left(z_1 - \frac{z_1 + z_2}{2}\right)^2} = \sqrt{\left(\frac{x_1 - x_2}{2}\right)^2 + \left(\frac{y_1 - y_2}{2}\right)^2 + \left(\frac{z_1 - z_2}{2}\right)^2} \\&= \sqrt{\frac{1}{4}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)} = \frac{1}{2}\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} = \frac{1}{2}d\end{aligned}$$

We see that the distance between P_2 & P_M is

$$\begin{aligned}d_M &= \sqrt{\left(x_2 - \frac{x_1 + x_2}{2}\right)^2 + \left(y_2 - \frac{y_1 + y_2}{2}\right)^2 + \left(z_2 - \frac{z_1 + z_2}{2}\right)^2} = \sqrt{\left(\frac{x_2 - x_1}{2}\right)^2 + \left(\frac{y_2 - y_1}{2}\right)^2 + \left(\frac{z_2 - z_1}{2}\right)^2} \\&= \sqrt{\frac{1}{4}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)} = \frac{1}{2}\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} = \frac{1}{2}d\end{aligned}$$

Therefore, we see that $P_M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}, \frac{z_1 + z_2}{2}\right)$ is indeed the midpoint between P_1 and P_2 .

8. Find an equation of a sphere if one of its diameters has endpoints $(2,1,4)$ and $(4,3,10)$.

We can determine the center of the sphere by finding the midpoint between these points. Therefore, the center of the sphere is $\left(\frac{4+2}{2}, \frac{3+1}{2}, \frac{10+4}{2}\right) = (3, 2, 7)$.

We then compute the radius by finding the distance between the center and one of our points.

$$r = \sqrt{(3-2)^2 + (2-1)^2 + (7-4)^2} = \sqrt{1+1+9} = \sqrt{11}.$$

Therefore, the equation of the sphere is $(x-3)^2 + (y-2)^2 + (z-7)^2 = 11$.

9. Find equations of spheres with center $(2, -3, 6)$ that touch (but do not intersect)

- a) the xy -plane

In order for the sphere to touch (and not intersect) the xy -plane the radius of the sphere must match the distance from the point $(2, -3, 6)$ to the xy -plane. This distance is 6 and so we let $r = 6$.

Therefore, the equation of the sphere is

$$(x-2)^2 + (y+3)^2 + (z-6)^2 = 36.$$

- b) the yz -plane

In order for the sphere to touch (and not intersect) the yz -plane the radius of the sphere must match the distance from the point $(2, -3, 6)$ to the yz -plane. This distance is 2 and so we let $r = 2$.

Therefore, the equation of the sphere is

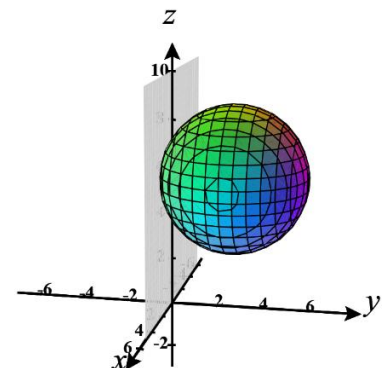
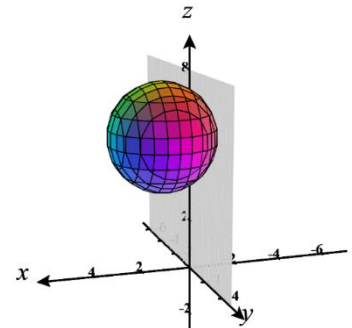
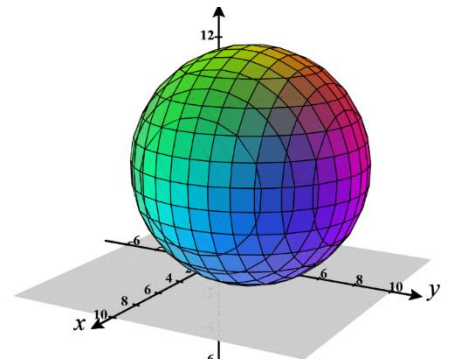
$$(x-2)^2 + (y+3)^2 + (z-6)^2 = 4.$$

- c) the xz -plane

In order for the sphere to touch (and not intersect) the xz -plane the radius of the sphere must match the distance from the point $(2, -3, 6)$ to the xz -plane. This distance is 3 and so we let $r = 3$.

Therefore, the equation of the sphere is

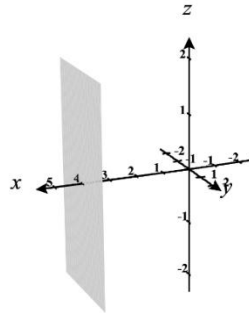
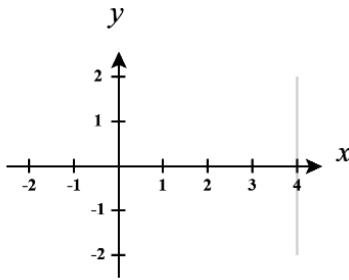
$$(x-2)^2 + (y+3)^2 + (z-6)^2 = 9.$$



10. What does the equation $x = 4$ represent in $\mathbb{R}^2$? What does it represent in $\mathbb{R}^3$? Provide sketches of each of these.

In $\mathbb{R}^2$ this represents a vertical line at $x = 4$.

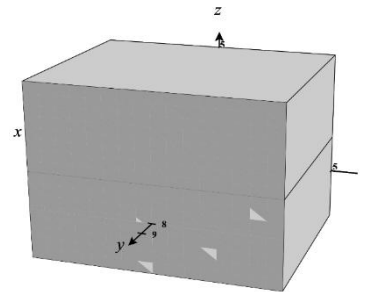
In $\mathbb{R}^3$ this represents a plane contain all points where $x = 4$. This plane runs parallel y -axis and the z -axis.



11. Describe in words the region of $\mathbb{R}^3$ represented by the following equations or inequalities.

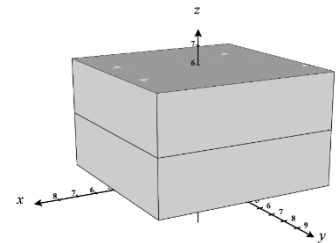
a) $y < 8$

This is the region of space where all y -coordinates are less than 8. This looks like a an infinitely sized rectangular prism.



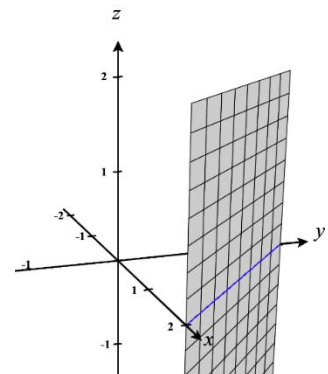
b) $0 \leq z \leq 6$

This is a region of space where all z -coordinates are between 0 and 6. This looks like a rectangular prism which extends infinitely in the x and y directions.



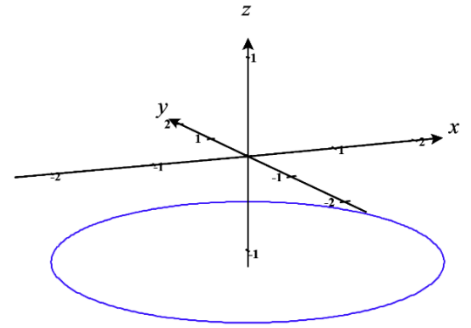
c) $x + y = 2$

This is a plane which runs parallel to the z -axis.



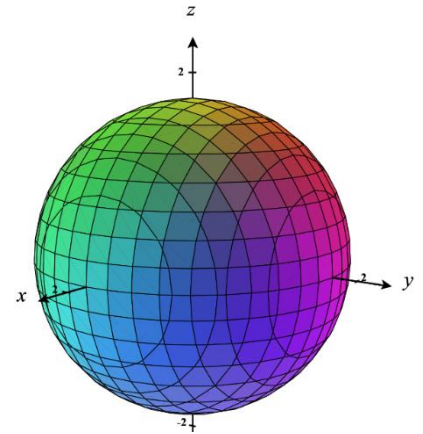
d) All points where $x^2 + y^2 = 4$ & $z = -1$

This creates a circle of radius 2 at the height of $z = -1$.



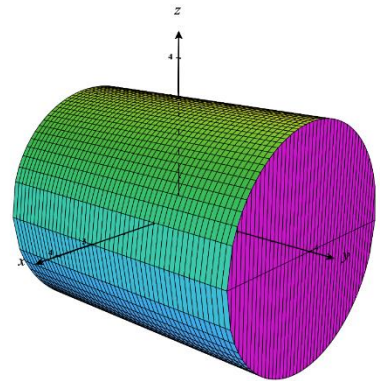
e) $x^2 + y^2 + z^2 \leq 3$

This is the sphere of radius $\sqrt{3}$ centered at the origin and also all points inside of this sphere.



f) $x^2 + z^2 \leq 9$

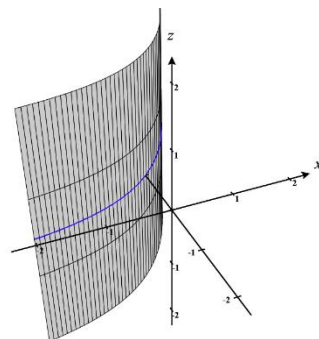
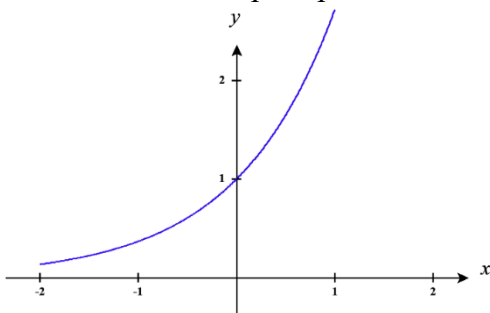
This is a cylinder with radius 3 and all points inside this cylinder. This cylinder runs along the y -axis.



12. Create a sketch of the following cylindrical surfaces in $\mathbb{R}^3$.

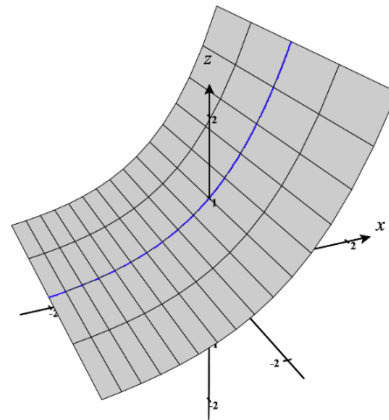
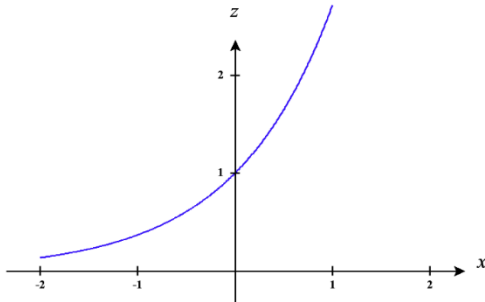
a) $y = e^x$

We start by sketching a picture of this curve in the xy -plane. Then since the z variable is free, we then extend this curve into 3-space parallel to the z -axis



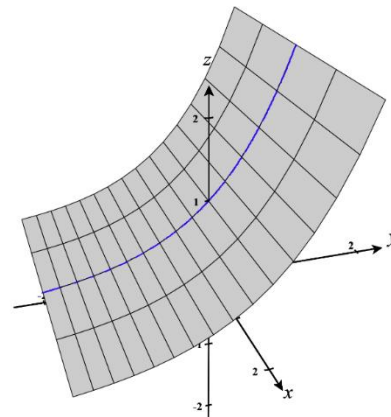
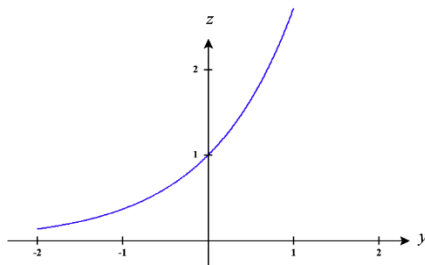
b) $z = e^x$

We start by sketching a picture of this curve in the xz -plane. Then since the y variable is free, we then extend this curve into 3-space parallel to the y -axis.



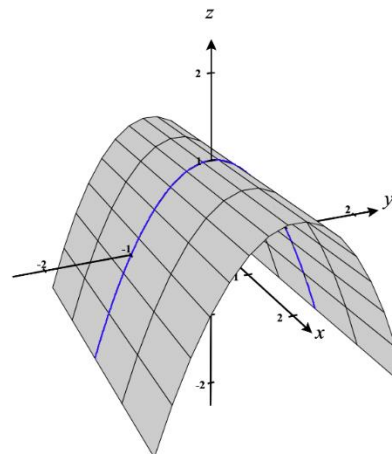
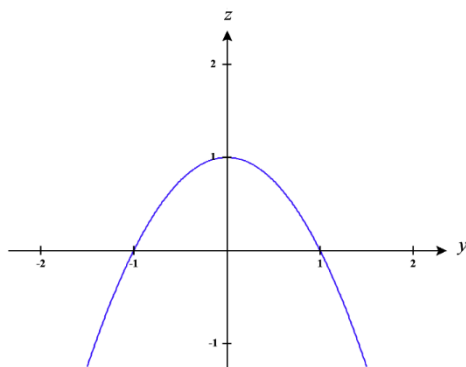
c) $z = e^y$

We start by sketching a picture of this curve in the yz -plane. Then since the x variable is free, we then extend this curve into 3-space parallel to the x -axis.



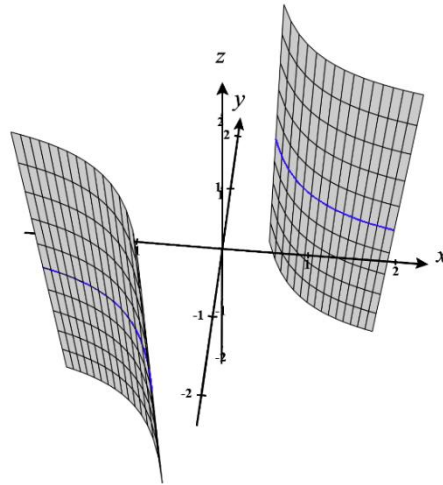
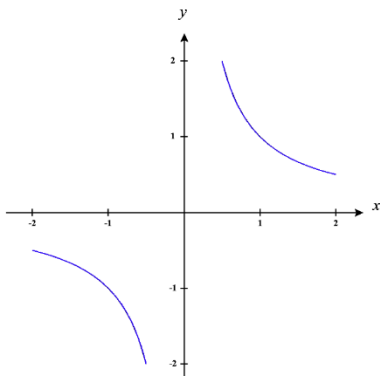
d) $z = 1 - y^2$

We start by sketching a picture of this curve in the yz -plane. Then since the x variable is free, we then extend this curve into 3-space parallel to the x -axis.



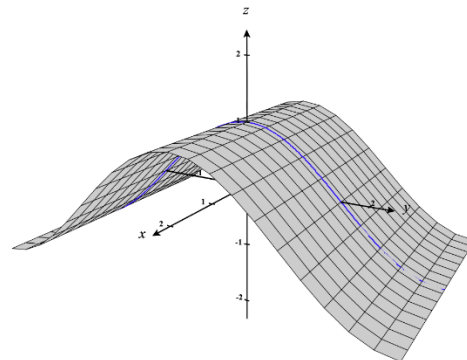
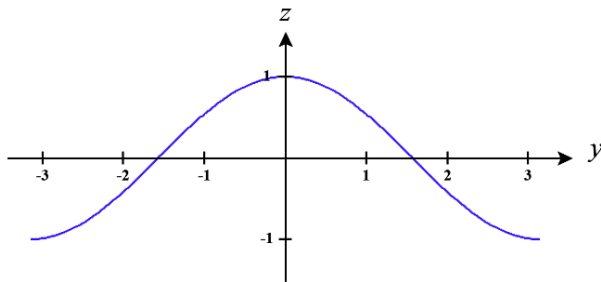
e) $xy = 1$

We start by sketching a picture of this curve in the xy -plane as $y = \frac{1}{x}$. Then since the z variable is free, we then extend this curve into 3-space parallel to the z -axis.



f) $z = \cos(y)$

We start by sketching a picture of this curve in the yz -plane. Then since the x variable is free, we then extend this curve into 3-space parallel to the x -axis.



g) $z = x + 1$

We start by sketching a picture of this curve in the xz -plane. Then since the y variable is free, we then extend this curve into 3-space parallel to the y -axis.

